

Q1/ what is pseudoscience? and how do you detect it?

A/

- pseudoscience is a fake science that makes claims based on faulty or nonexistent scientific evidence. In most cases, these pseudosciences present claims in a way that makes them seem possible, but with little or no empirical support for these claims.

Graphology, numerology, and astrology, are all examples of pseudosciences. In many cases, these pseudosciences rely on anecdotes and testimonials to back up their often outlandish claims.

- If you are trying to determine if something is a pseudoscience, there are a few key things you can look for:

Consider the purpose. Science is focused on helping people develop a deeper, richer, and fuller understanding of the world.

Pseudoscience often focuses on furthering some type of ideological agenda.

Consider how challenges are dealt with. Science welcomes challenges and attempts to disprove or refute different ideas. Pseudoscience, on the other hand, tends to greet any challenges to its dogma with hostility.

Look at the research. Science is supported by a deep and ever-growing body of knowledge and research. Ideas around the topic may have changed over time as new things are discovered and new research is performed. Pseudoscience tends to be fairly static. Little may have changed since the idea was first introduced and new research might not exist.

Can it be proven false? Falsifiability is a key hallmark of science. This means that if something is false, researchers could prove that it was false. Many pseudoscientific claims are simply untestable, so there is no way for researchers to prove these claims false.

(ممکن تختصرون هذا الجواب اكثر ^)

Q2/

Q2: Distinguish the credible sources of information in the following paragraph: (10 marks)

The Internet of Things (IoT) is a novel technology that allows devices to create a global communication network by exchanging data through the internet and acting on that data [1]. Although IoT is a relatively new concept to most, the interconnective technology has been around for two decades [2].

Kevin Ashton of MIT first used the term "Internet of Things" in 1999 [3]. From there on, LG introduced the first-ever smart fridge to the world in 2000, and seven years later, the first iPhone was released [4].

Since devices have multiple sensors, they can acquire considerable data from numerous sources, giving them more information to work with when acting on data received [5].

A big reason for the invention of IoT is convenience. Smart devices that automate daily tasks allow humans to do other activities. These devices ultimately lighten people's workload [6].

Another example is a self-driving car, connecting to the Internet to find the quickest route to a destination [7]. This is the ultimate convenience for humans. The room for innovation within IoT is massive.

Web-based devices save people money and time. This includes planning work schedules, time tracking, effective communication, and setting reminders for daily tasks [8].

Having IoT devices track and order things for you, turn lights off automatically when you leave the room, and manage tasks for which you don't have time is the ultimate convenience! [9].

More and more of these devices will become available for use over the coming years, with an estimated total number of IoT connections to reach 27 billion in 2024 [10].

References

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- [10] Bhojar, Prachin, Sahare, Parul, Dhok, S.B., and R.B. Deshmukh. "Communication technologies and security challenges for internet of things: A comprehensive review." AEU - International Journal of Electronics and Communications 99, (2019): 81-99. Accessed December 4, 2022. <https://doi.org/10.1016/j.aue.2018.11.031>.

A/

[1] invalid.

[2] – [10] valid

هنا التقييم يعتمد حسب وجهة نظرك، عدا اول وحدة لانه مستخدم ويكيديا فتلقائيا خطأ، الباقيات تحددن اذا مقبولات او لا حسب اذا كانت المعلومة تحتاج مصدر او كمية معلومات كافية لهالشي او اي شي من هالقبيل لكن المصادر بحد ذاتها ما بيها شي.

انا ما شفت بيهن هيچ اشكال فمشيتهن، ممكن نعتبر مصدر 3 غير مقبول لانه المعلومة كلش مختصرة؟ بس مو بالضرورة. بكيفكم التقييم هنا.

Q3/ Define research methods and methodology. What is the major difference between them?

A/

- research methods (scientific methods): a method of procedure consisting of systematic observation, measurement, and experiment, and the formulation, testing, and modification of hypotheses.

- Methodology:

Method (*Also called Materials and Methods or Experimental Methods*)

This section provides the readers with enough detail that they can replicate your research. Explain how you studied the problem, identify the procedures you followed, and structure these as logically as possible. If your methods are new, you will need to explain them in detail. If they have been published before, name the method and cite the previously published work; if you have modified the methods, refer to the original work and include your amendments. Identify the equipment and describe materials used and specify the source if there is variation in quality of materials. Include the frequency of observations, what types of data were recorded. Be precise in describing measurements and include strengths and weaknesses of measurement. Name any statistical tests used so that your numerical results can be validated. It is advisable to use the past tense, and avoid using the first person.

ممکن تختصرون التعریف هنا^

- the methodology is part of the scientific paper structure, and the research methods are the steps that the researcher commits in order to prepare the scientific paper including its methodology section.

Q4/ How do you use the steps of the scientific methods to prepare a scientific article?

A/

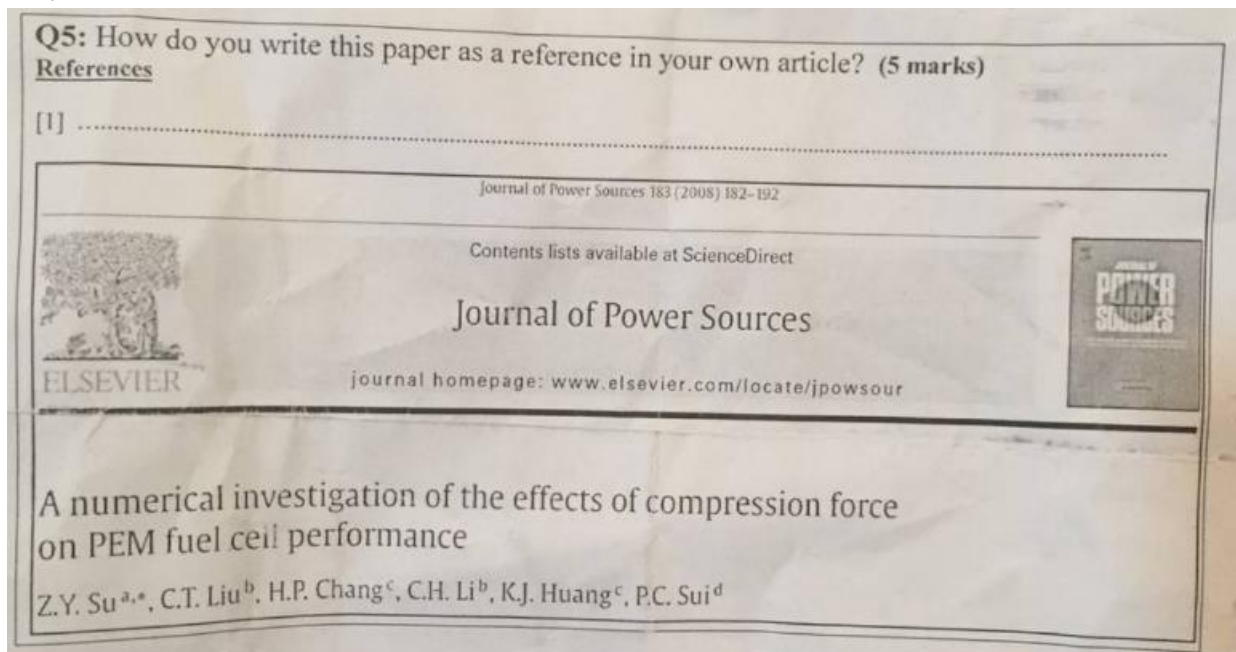
Scientific Method Reviewed

هنا هاي الخطوات بس تعداد, الشرح
مالتين كله بملف ال

3.scientific methods

- ASK Question
- Do Background Research
 - Construct Hypothesis
 - Test with an Experiment
- Analyze Results and Draw Conclusions
 - Hypothesis is True
 - If: Hypothesis is False (Go back to Step 3)
 - Report Results

Q5/



A/

[1] Z.Y.Su, C.T.Liu, H.P.Chang, C.H.Li, K.J.Huang, P.C.Sui. A numerical investigation of the effects of compression force on PEM fuel cell performance. *Journal of Power Sources*. 183(2008), 182-192. www.elsevier.com/locate/jpowsour

ترتيب الجواب كان مثل الي موجود بملف

4. Researching a topic, evaluating information, and Literature survey

Author 1, Author 2, ..., ... Title. Journal. Volume (Year), pp.## - ##.

**Luigi Ventola, Francesco Robotti, Masoud
Dialameh, Flaviana Calignano, Diego
Manfredi, Eliodoro Chiavazzo, Pietro Asinari.
Rough surfaces with enhanced heat transfer
for electronics cooling by direct metal laser
sintering. International Journal of Heat and
Mass Transfer. 75 (2014), pp.58 - 74.**

بالنسبة للرابط الي بالنهاية هو رابط للمجلة عامة وليس المقالة نفسها فمو بالضرورة تضيفونه, لكن الروابط
توضع بنهاية المصدر في حال الرغبة باضافتها.